DISTINCT RUNS IN STANDARD-ORDER COMPOSITIONS: A LENGTH FORMULA AND NEW TERMS OF OEIS A351015

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Abstract. Let $a(n)$ be the smallest nonnegative integer k such that the k-th composition in standard order has exactly n distinct runs; this is OEIS A351015. A run consisting of l copies of a part v is represented by the type (v, l), and we observe that its contribution to the length of the standard binary encoding is vl. This yields an exact formula for the binary length of $a(n)$: it equals the sum of the first n terms of OEIS A061017, the positive integer multiplication table sorted with multiplicity. We prove that the resulting lower bound is always attainable, give an exact lexicographic dynamic program for $a(n)$, and obtain

$$a(10) = 4\,503\,885\,791.$$

The computation supplies terms through $a(16)$, while the length formula implies $\log_2 a(n) \sim n^2/(2\log n)$.

1. Introduction

The standard order on integer compositions gives a bijection between the nonnegative integers and the finite compositions. If the binary expansion of $k > 0$ is read from left to right, each 1 starts a new part of size 1 and each subsequent 0 increases the current part by 1. Equivalently, a part v is encoded by the word 10^{v-1}, and the integer 0 corresponds to the empty composition. This encoding is OEIS A066099 [2].

A *run* is a maximal consecutive block of equal parts. Thus a run of l copies of the value v has type (v, l), and two runs are distinct exactly when their types differ. OEIS A351015, introduced by Gus Wiseman [1], is

$$a(n) = \min\{k \geq 0 : \text{ the standard composition of } k \text{ has } n \text{ distinct runs}\}.$$

The original entry listed $a(0)$ through $a(7)$. The terms

$$a(8) = 4\,617\,711, \qquad a(9) = 138\,835\,439$$

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were contributed by the author and approved by the OEIS on June 28, 2026. The purpose of this note is to replace a linear scan over integers by a structural description together with an exact finite optimization.

Runs in integer compositions have previously been studied through generating functions for restrictions on run length; see Wilf [9]. A close OEIS analogue is A350952 [5], which asks for the smallest integer whose binary expansion has a prescribed number of distinct runs. That problem has a two-letter alphabet and charges one bit per position. Here the part values are unbounded, and a run of length l in the value v has cost vl, producing a different extremal problem.

Several neighbouring OEIS sequences concern the same statistic on standard-order compositions. A351014 [7] gives the number of distinct runs of the n-th composition, of which A351015 is the sequence of first appearances; A351013 [6] counts the compositions of n whose runs are all distinct, and A351290 [8] lists the standard indices of such compositions. The related statistical question of the longest single run in a random composition is treated by Gafni [10]. None of these addresses the minimum-size (first-occurrence) problem or the length formula obtained here.

Our starting point is the elementary but decisive observation that the standard encoding turns a purely combinatorial constraint (the number of distinct runs) into an additive one (the binary length). Section 2 makes this precise and proves that the induced lower bound on bitlength$(a(n))$ is always attained. Section 3 refines the length formula into an exact lexicographic dynamic program that pins down the least integer of that length. Section 4 records the resulting values through $a(16)$ and gives a self-contained certificate for $a(10)$, and Section 5 derives the asymptotic growth rate.

2. The minimum possible binary length

Let $s(1), s(2), \ldots$ denote OEIS A061017 [3]:

$$1, 2, 2, 3, 3, 4, 4, 4, 5, 5, 6, 6, 6, 6, 7, 7, \ldots.$$

Equivalently, each positive integer m occurs $d(m)$ times, where $d(m)$ is the number of positive divisors of m. This is precisely the multiset of products vl over ordered positive pairs (v, l), sorted into nondecreasing order. Set

$$L(0) = 0, \qquad L(n) = \sum_{j=1}^{n} s(j) \quad (n \geq 1).$$

Lemma 2.1 (Rearrangement)**.** *A multiset of $N \geq 1$ labels can be placed in a linear order with no two equal labels adjacent if and only if its largest multiplicity M satisfies $M \leq N - M + 1$.*

Proof. The M copies of a most frequent label require at least $M - 1$ separating labels, so $M - 1 \leq N - M$; this proves necessity. For sufficiency we induct on N, the cases $N \leq 2$ being immediate. Let $N \geq 3$ with $2M \leq N+1$. Then at least two labels have positive multiplicity, since a single label would

give $M = N$ and hence $2N \le N + 1$, i.e. $N \le 1$. Let x and y be labels of largest and second-largest positive multiplicity, and let c be the third-largest multiplicity, with $c = 0$ if fewer than three labels occur. Delete one copy of each of x and y. In the remaining multiset of $N - 2$ labels the largest multiplicity is $M' = \max(M - 1, c)$. If $M' = M - 1$ then $2M' = 2M - 2 \le N - 1$. If $M' = c$, then $c \le M$, with $c = M$ only when at least three labels attain M, whence $N \ge 3M$ and $2c = 2M \le N - M \le N - 1$; and if $c \le M - 1$ then $2c \le 2M - 2 \le N - 1$. In every case $2M' \le N - 1 = (N - 2) + 1$, so the induction hypothesis provides an arrangement w of the $N - 2$ remaining labels with no two equal adjacent. Write x and y before w, choosing which one sits immediately before w so that it differs from the first label of w; this is possible because $x \ne y$, so at most one of them equals that label. The two new adjacencies are then x–y and (a label)–w_1, both between distinct labels, so the whole word has no two equal adjacent labels. □

Theorem 2.2. *For every $n \ge 0$,*

$$\operatorname{bitlength}\big(a(n)\big) = L(n) = \sum_{j=1}^{n} s(j),$$

where $\operatorname{bitlength}(0) = 0$.

Proof. The case $n = 0$ is immediate, so assume $n \ge 1$. A run of type (v, l) contains l parts of size v; it therefore contributes vl both to the sum of the composition and to the length of its standard binary encoding. Any composition with n distinct run types must contain at least one copy of each type, so its encoding length is at least the sum of the n smallest elements of the multiset of type costs, namely $L(n)$. This establishes the lower bound.

It remains to construct a composition attaining it. For $q \ge 0$, let

$$D(q) = \sum_{m \le q} d(m) = \#\{(v, l) \in \mathbb{Z}_{>0}^2 : \ vl \le q\},$$

the divisor summatory function (OEIS A006218 [4]). Choose q so that $D(q-1) < n \le D(q)$, and put $r = n - D(q-1)$. Take every run type (v, l) with $vl < q$, together with r run types satisfying $vl = q$: if $r < d(q)$, choose any r boundary types other than $(1, q)$; if $r = d(q)$, take all boundary types.

We show that these types can be ordered so that consecutive values v differ, applying Lemma 2.1 with each type's value v as its label. Let $N = n$ be the number of selected types and M the number of them with value 1. Take $q \ge 3$; the cases $q = 1, 2$ are immediate.

If $(1, q)$ is omitted, then $M = q - 1$, while each value $v \ge 2$ is carried by at most $\lfloor (q-1)/v \rfloor + 1 \le q - 1$ selected types, so M is the largest label multiplicity. The $2q - 3$ pairs $(1, l)$ for $1 \le l \le q - 1$ and $(v, 1)$ for $2 \le v \le q - 1$ are distinct with product $< q$, so $D(q-1) \ge 2q - 3$; hence $N = n > D(q-1) \ge 2q - 3$, which gives $2M = 2(q-1) \le N + 1$, that is, $M \le N - M + 1$. If instead all boundary types are selected, then $M = q$ and $N = n = D(q)$, while each value $v \ge 2$ occurs in $\lfloor q/v \rfloor \le q = M$ types,

so M is again largest; adjoining $(1, q)$ and $(q, 1)$ to the same witnesses gives $D(q) \geq 2q - 1$, so $2M = 2q \leq N + 1$. In both cases Lemma 2.1 yields the required ordering.

Order the value labels with no equal adjacent labels and assign the selected multiplicities l to their corresponding positions. Adjacent run types now have different values, so their concatenation is a valid composition with exactly n distinct run types and total cost $L(n)$. The lower bound is therefore attained. □

An explicit arithmetic form of the theorem is useful in the sequel. If $D(q-1) < n \leq D(q)$, then

$$L(n) = \sum_{m<q} m\, d(m) + q\big(n - D(q-1)\big). \tag{2.1}$$

3. An exact algorithm for $a(n)$

Theorem 2.2 fixes the minimum word length, but not the least word of that length. All minimum-cost type sets contain every (v, l) with $vl < q$ and differ only in the choice of r divisors on the boundary $vl = q$. Equality in the lower bound also forces every selected run type to occur exactly once, since any additional occurrence has positive cost and would make the word longer than $L(n)$.

For a run type (v, l), write

$$B(v, l) = \big(10^{v-1}\big)^{l}$$

for its binary block. Fix a candidate type set T. If $S \subseteq T$ is the set of *unused* types and u is the value of the preceding run—with $u = 0$ before the first run, a sentinel distinct from every value—define

$$F(S, u) = \min_{\substack{(v,l)\in S \\ v\neq u}} B(v, l) \cdot F\big(S \setminus \{(v, l)\}, v\big),$$

where the product denotes concatenation and the minimum is lexicographic. The base case is $F(\varnothing, u) = \varepsilon$, the empty word, and a state with no admissible transition is infeasible, with value $+\infty$ (larger than every finite word) so that it never attains a minimum. The least valid concatenation of T is $F(T, 0)$. Taking the further minimum of $F(T, 0)$ over the admissible boundary choices of T yields the least binary word of length $L(n)$, hence the least integer $a(n)$. For a fixed candidate set the dynamic program has $O(n\, 2^n)$ states, with up to n transitions each: exponential in n, but dramatically smaller than scanning all integers up to $a(n)$.

4. New terms and a certificate for $a(10)$

The exact computation gives the values in Table 1. The entries through $a(9)$ agree with OEIS A351015.

TABLE 1. The minimum binary length $L(n)$ and the term $a(n)$ of OEIS A351015, for $0 \le n \le 16$.

n	$L(n)$	$a(n)$
0	0	0
1	1	1
2	3	5
3	5	27
4	8	155
5	11	1 655
6	15	18 039
7	19	281 975
8	23	4 617 711
9	28	138 835 439
10	33	4 503 885 791
11	39	279 381 792 735
12	45	17 876 500 338 655
13	51	1 144 081 370 306 527
14	57	73 222 145 385 127 871
15	64	9 296 594 182 239 903 679
16	71	1 189 966 115 078 053 658 495

For $n = 10$, the ten smallest run-type costs are

$$1, 2, 2, 3, 3, 4, 4, 4, 5, 5,$$

whose sum is 33. Equality forces precisely the ten types (v, l) with $vl \le 5$. The dynamic program returns the binary word

$$\texttt{10000110001110011111010111111011111},$$

which encodes the composition

$$(5, 1, 4, 1, 1, 3, 1, 1, 1, 2, 2, 1, 1, 1, 1, 2, 1, 1, 1, 1, 1).$$

Its run types, in order, are

$$(5, 1), (1, 1), (4, 1), (1, 2), (3, 1), (1, 3), (2, 2), (1, 4), (2, 1), (1, 5).$$

They are all distinct, their total cost is 33, and exhaustive dynamic programming over every valid ordering proves that the displayed word is the least word of that length.

There is also a short direct argument. Write $A_j = (1, j)$ for $1 \le j \le 5$; the other five types are $(5, 1), (4, 1), (3, 1), (2, 2), (2, 1)$, of which only $(2, 2)$ and $(2, 1)$ share a value. A least word cannot begin with an A_j, because that gives the prefix `11`, whereas every other type begins `10`. Since no two value-1 types may be adjacent, the five A_j occupy one each of the five gaps after the other blocks, and the word alternates non-A, A, non-A, ..., ending in an A_j.

Greedy lexicographic minimization fixes both orders. Among the non-A blocks, $(5,1) = \texttt{10000}$, $(4,1) = \texttt{1000}$, and $(3,1) = \texttt{100}$ have a 0 in their third bit, whereas $(2,2) = \texttt{1010}$ and $(2,1) = \texttt{10}$ (followed by the 1 of the next block) have a 1 there; so the three value-≥ 3 blocks come first, and among them the longest—hence zero-richest—first, giving $(5,1), (4,1), (3,1)$. The two value-2 types are then compared at the fourth bit of their common suffix: placing $(2,2)$ before $(2,1)$ writes $\texttt{1010}$, while the reverse writes $\texttt{10}$ followed by an all-ones block A_j, giving prefix $\texttt{1011}$ (if $A_j = A_1$ is a single $\texttt{1}$, the fourth bit is instead the $\texttt{1}$ beginning the next block, so the prefix is still $\texttt{1011}$); since $\texttt{1010} < \texttt{1011}$, the order is $(2,2), (2,1)$. At each of the first four A slots the smallest unused j reaches the following 0 soonest, placing A_1, A_2, A_3, A_4; the remaining A_5 fills the final slot. The resulting word is the one displayed above, and it is the unique lexicographic minimum, certifying $a(10) = 4\,503\,885\,791$ without a scan of smaller integers.

5. Asymptotic growth

Here and below, log denotes the natural logarithm. The classical divisor summatory estimate [11, §18.2] is

$$D(x) = x \log x + (2\gamma - 1)x + O(\sqrt{x}),$$

and partial summation gives

$$\sum_{m \leq x} m\, d(m) = \tfrac{1}{2} x^2 \log x + O(x^2).$$

If $q = s(n)$, then $n \sim q \log q$, so $q \sim n/\log n$. Writing $r = n - D(q-1)$, Equation (2.1) has boundary contribution qr, where $1 \leq r \leq d(q) \leq q$. Hence $qr = O(q^2) = o(q^2 \log q)$, and therefore (2.1) yields

$$L(n) \sim \frac{n^2}{2 \log n}.$$

For $n \geq 1$, since $2^{L(n)-1} \leq a(n) < 2^{L(n)}$, we obtain the following consequence.

Corollary 5.1.

$$\log_2 a(n) \sim \frac{n^2}{2 \log n}.$$

6. Reproducibility

The accompanying file `a351015_exact.py` uses only the Python standard library. It implements the boundary enumeration and the exact bitmask dynamic program, checks the structural certificate for each returned term, and verifies all recorded values through $a(16)$. The full run is invoked by

```
python a351015_exact.py --max-n 16
```

and adding `--details` prints the binary words and run-type orderings.

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